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MH1200 Linear Algebra I

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1. Elementary Matrix

Definition Elementary Matrix

Definition 1

A **square** matrix E is elementary iff we can obtain E using exactly one elementary row operation on I.

Note that I itself is elementary since multiplying a row by 1 is an elementary row operation.

The main reason why elementary matrices are useful is because they **perform elementary row operations for you.**

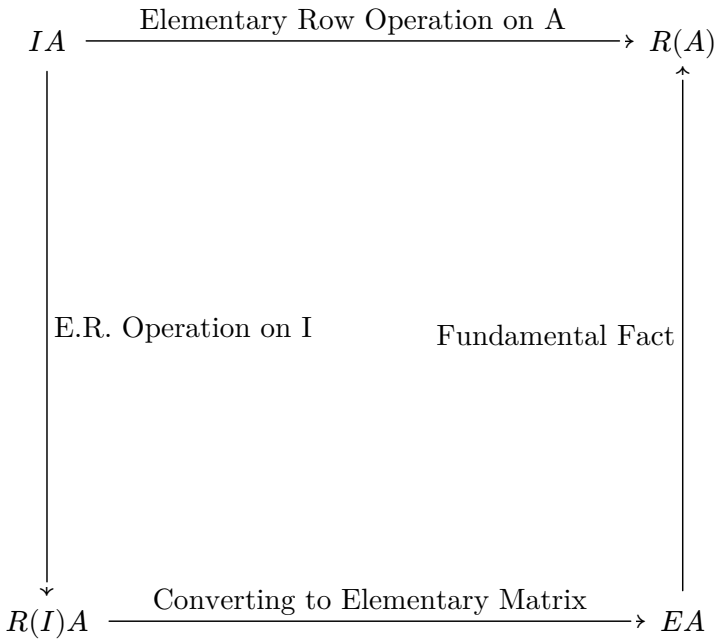
1.1. The Fundamental Fact of Elementary Matrices

Let $R(A)$ be the result of an elementary row operation on A. Then, there exists an elementary matrix E such that:

$$EA = R(A)$$

Note: Elementary matrices are **always** square. This is because they must preserve the dimensions of the input matrix.

1.2. Diagram



Left multiplication by an elementary matrix represents the group action of performing a row operation.

1.3. Example

Consider:

$$E_1 E_2 A$$

With:

$$E_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Notice that E_2 corresponds to $R_2 \leftarrow R_2 + R_4$, while E_1 corresponds to $R_2 \leftrightarrow R_4$.

Therefore, $E_1 E_2 A$ is the result of doing the following to A, in order:
1. $R_2 \leftarrow R_2 + R_4$
2. $R_2 \leftrightarrow R_4$

1.4. Another Consequence of the Fundamental Fact

Example Composition of Elementary Row Operations

Example 2

Consider a list of k elementary row operations:
- Row operation 1, with corresponding elementary matrix E_1,

$$\vdots$$

- Row operation k, with corresponding elementary matrix E_k.

Suppose we perform these elementary row operations, in order, to A. Express the resulting matrix as a matrix product.

Solution Composition of Elementary Row Operations

Solution 3

$$A$$
$$\downarrow$$
$$E_1 A$$
$$\downarrow$$
$$E_2(E_1 A)$$
$$\downarrow$$
$$E_3(E_2(E_1 A))$$
$$\downarrow$$
$$\vdots$$
$$\downarrow$$
$$E_k E_{k-1} E_{k-2} \cdots E_1 A$$

1.5. Usage of Π in Matrix Products

Avoid using Π for matrix products. Recall that matrix multiplication is not commutative.

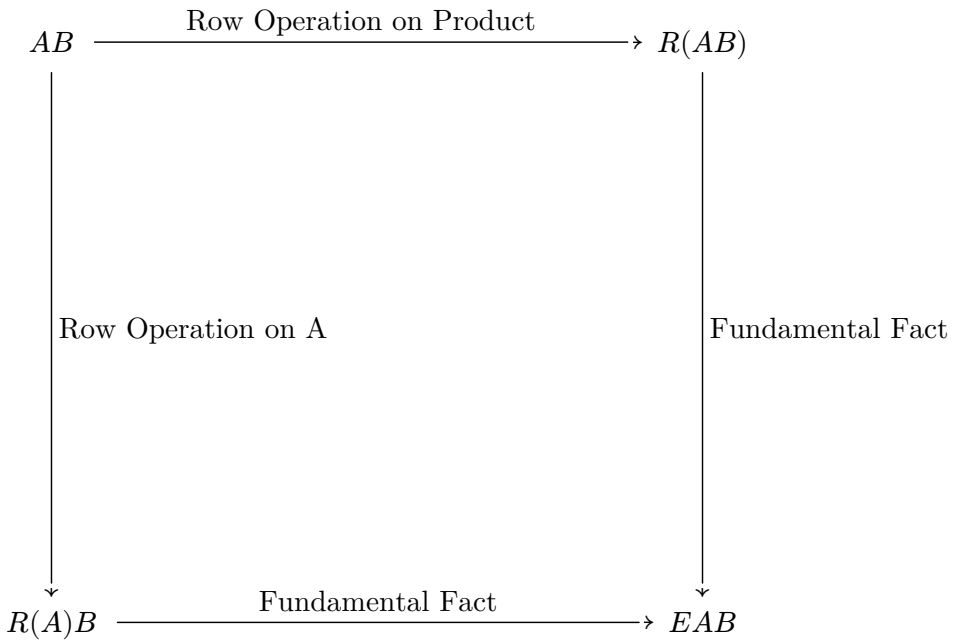
1.6. Why is the Fundamental Fact True?

Lemma Row Operations Respect Matrix Multiplication

Lemma 4

Let $R(A)$ denote the result of applying some chosen row opeartion R to some matrix. Then: $R(AB) = R(A)B$

Note: A lemma is a "less important" theorem that's mostly used to prove a bigger theorem.



Proof | Row Operations Respect Matrix Multiplication

Proof 5

To prove this, we will do casework on all possible row operations.

Row Operation 1: $R_x \to cR_x$

Notice that:

$$\begin{aligned}
(R(A)B)_{ij} &= \sum_{p=1}^{n} R(A)_{ip} B_{pj} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x \\ \sum_{p=1}^{n} (cA_{ip}) B_{pj} \text{ if } i = x \end{cases} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x \\ c \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i = x \end{cases} \\
&= \begin{cases} (AB)_{ij} \text{ if } i \neq x \\ c(AB)_{ij} \text{ if } i = x \end{cases} \\
&= R(AB)_{ij}
\end{aligned}$$

Row Operation 2: $R_x \to R_x + cR_y$

Notice that:

$$\begin{aligned}
(R(A)B)_{ij} &= \sum_{p=1}^{n} R(A)_{ip} B_{pj} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x \\ \sum_{p=1}^{n} (A_{ip} + cA_{yp}) B_{pj} \text{ if } i = x \end{cases} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x \\ \sum_{p=1}^{n} A_{ip} B_{pj} + cA_{yp} B_{pj} \text{ if } i = x \end{cases} \\
&= \begin{cases} (AB)_{ij} \text{ if } i \neq x \\ (AB)_{ij} + c(AB)_{yj} \text{ if } i = x \end{cases} \\
&= R(AB)_{ij}
\end{aligned}$$

Row Operation 3: $R_x \leftrightarrow R_y$

Notice that:

$$\begin{aligned}
(R(A)B)_{ij} &= \sum_{p=1}^{n} R(A)_{ip} B_{pj} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x, y \\ \sum_{p=1}^{n} (A_{yp}) B_{pj} \text{ if } i = x \\ \sum_{p=1}^{n} (A_{xp}) B_{pj} \text{ if } i = y \end{cases} \\
&= \begin{cases} \sum_{p=1}^{n} A_{ip} B_{pj} \text{ if } i \neq x, y \\ \sum_{p=1}^{n} A_{yp} B_{pj} \text{ if } i = x \\ \sum_{p=1}^{n} A_{xp} B_{pj} \text{ if } i = y \end{cases} \\
&= \begin{cases} (AB)_{ij} \text{ if } i \neq x, y \\ (AB)_{yj} \text{ if } i = x \\ (AB)_{xj} \text{ if } i = y \end{cases} \\
&= R(AB)_{ij}
\end{aligned}$$

Since we have exhausted all cases, we have:

$$(R(A)B)_{ij} = R(AB)_{ij}$$

Thus,

$$R(A)B = R(AB)$$

As needed. ■

Corollary 7

Corollary Multiplication by Elementary Matrices Corresponds to Row Operations

The result of an elementary row operation is the same as multiplying by the corresponding elementary matrix. That is:

$$R(A) = EA$$

Proof Multiplication by Elementary Matrices Corresponds to Row Operations

Proof 8

$$\begin{aligned} R(A) &= R(IA) \\ &= R(I)A \\ &= EA \end{aligned}$$

1.7. Invertibility of Elementary Matrices

Theorem Invertibility of Elementary Matrices

Theorem 9

All elementary matrices E are invertible.

Proof Invertibility of Elementary Matrices

Proof 10

Let E be an elementary matrix. Then, there is some row operation so that:

$$E = R(I)$$

Let us do casework on the elementary row operations:

Case 1: $R_x \to cR_x$

Notice that $R_x \to \frac{1}{c}R_x$ undoes this operation. Let $R_2(A)$ represent applying this row operation on a matrix A, and let E' be the corresponding elementary matrix.

Then:

$$\begin{aligned} I &= R_2(R(I)) \\ &= R_2(EI) \\ &= E'EI \\ &= E'E \end{aligned}$$

So, E' is the left inverse of E, and is thus the inverse of E.

Note: We must also show that $I = EE'$, but this is also simple. The key identity here is $R(R_2(I))$. The idea is similar and will thus be skipped for now. Note that you can also prove that left inverses are also right inverses.

Case 2: $R_x \to R_x + cR_y$

Notice that $R_x \to R_x - cR_y$ undoes this operation. The proof in this case is similar to case 1.

Case 3: $R_x \leftrightarrow R_y$

Notice that $R_x \leftrightarrow R_y$ undoes this operation. The proof is similar to case 1.

Since we have exhausted all cases, E is invertible, as needed. ■

Corollary Inverses of Elementary Matrices are Elementary

Corollary 12

The inverse E^{-1} of an elementary matrix E is elementary.

Proof Inverses of Elementary Matrices are Elementary

Proof 13

Notice in the proof above that the operation used to undo every elementary row operation is also an elementary row operation. Therefore, the corresponding inverse is also an elementary matrix.